

**ForLens – A Lightweight AI-Driven Endpoint Security Solution
for Real-Time Threat Detection in SMEs**

25-26J-076

Final Report - Draft

Salgado M. B. D

B.Sc. (Hons) Degree in Information Technology Specialized in Cyber
Security

Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Salgado M. B. D	IT21289934	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Insider threats remain one of the most critical and challenging cybersecurity issues, particularly for Small and Medium Enterprises (SMEs), where limited resources restrict the deployment of advanced security solutions. Traditional systems such as Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) rely heavily on rule-based detection and predefined signatures, making them ineffective against evolving and behavior-based threats. Furthermore, these systems often lack explainability, generate high false positives, and require significant infrastructure and expertise.

This research presents ForLens, a lightweight AI-driven endpoint security and forensic framework designed to detect threats in real time using behavioral analytics, machine learning, and integrated threat intelligence. The proposed system combines an unsupervised anomaly detection pipeline, utilizing models such as Isolation Forest and Autoencoders, with a custom-built ForLens Threat Intelligence (FTI) engine that aggregates and normalizes data from open-source intelligence sources including URLHaus, ThreatFOX and MalwareBazaar. This integration enables the system to correlate internal behavioral anomalies with external Indicators of Compromise (IoCs), enhancing detection accuracy and contextual awareness.

Additionally, ForLens incorporates a lightweight endpoint agent for continuous monitoring of user and system activities, along with a cryptographically secured forensic logging mechanism that ensures data integrity and non-repudiation through hash chaining techniques. The system further employs explainable AI techniques such as SHAP and LIME to provide transparent, human-readable justifications for each detected anomaly, improving analyst trust and decision-making.

Experimental evaluation demonstrates that the proposed system achieves high detection accuracy while maintaining low computational overhead, making it suitable for SME environments. The integration of behavioral machine learning, threat intelligence enrichment, and forensic readiness provides a scalable and cost-effective solution for proactive insider threat detection.

This research contributes to both academic and practical domains by presenting a reproducible, explainable, and lightweight cybersecurity framework tailored for modern enterprise environments.

Keywords: Insider Threat Detection, Machine Learning, Threat Intelligence, Anomaly Detection, Digital Forensics, Explainable AI

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CTI	Cyber Threat Intelligence
EDR	Endpoint Detection and Response
FTI	ForLens Threat Intelligence
IDS	Intrusion Detection System
IOC	Indicator of Compromise
IoC DB	Indicator of Compromise Database
ML	Machine Learning
OSINT	Open-Source Intelligence
SIEM	Security Information and Event Management
SME	Small and Medium Enterprise
SOAR	Security Orchestration, Automation and Response
XDR	Extended Detection and Response

1. Introduction

Insider threats continue to represent one of the most persistent and complex challenges in modern cybersecurity. Unlike external attacks, insider threats originate from individuals who possess legitimate access to organizational systems, enabling them to bypass traditional security controls and operate within trusted environments. This inherent advantage makes insider activities significantly harder to detect, as malicious actions often resemble normal user behavior. Industry evidence indicates that insider-related incidents contribute to an increasing proportion of data breaches, frequently resulting in greater financial and reputational damage due to delayed detection and response mechanisms [1].

Conventional enterprise security solutions, including Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) systems, have been widely adopted to monitor and analyze security events. However, these systems primarily rely on rule-based detection, signature matching, and predefined policies. While effective for identifying known threats, such approaches are limited in detecting previously unseen or evolving attack patterns. In particular, they struggle to capture subtle behavioral deviations such as abnormal login times, privilege misuse, or gradual data exfiltration. Furthermore, rule-based systems often generate high volumes of false positives and require significant computational resources and expert configuration, making them less suitable for Small and Medium Enterprises (SMEs) [2].

Recent advancements in artificial intelligence and machine learning have introduced more adaptive approaches for detecting malicious behavior. Unsupervised anomaly detection techniques, such as Isolation Forest, Autoencoders, and Long Short-Term Memory (LSTM) networks, enable systems to model baseline user behavior and identify deviations without requiring labeled datasets. These methods are particularly effective in insider threat detection scenarios, where labeled malicious data is scarce. By learning normal behavioral patterns from system logs and user activities, machine learning models can dynamically detect anomalies and adapt to evolving threat landscapes [3].

Despite these advantages, anomaly detection models alone lack contextual awareness, which is essential for accurate threat identification. Not all anomalies correspond to malicious activity; therefore, relying solely on behavioral deviations may lead to increased false positives. To address this limitation, the integration of Cyber Threat Intelligence (CTI) has emerged as a critical enhancement in modern detection systems.

In this research, threat intelligence is specifically derived from open-source platforms such as URLHaus, ThreatFox, and MalwareBazaar, which provide real-time indicators of compromise including malicious URLs, IP addresses, file hashes, and malware signatures. These platforms enable the enrichment of detected anomalies by correlating internal behavioral patterns with externally verified threat indicators. Such integration improves detection accuracy, reduces false positives, and enhances the ability to identify real-world attack patterns. [4]

Building upon these advancements, the ForLens project proposes a lightweight and explainable AI-driven forensic detection framework tailored for SME environments. The system integrates behavior-aware anomaly detection with a custom-built Threat Intelligence module that aggregates and normalizes data from URLHaus, ThreatFox, and MalwareBazaar into a unified enrichment pipeline. This hybrid approach enables the system to combine internal behavioral

analysis with external threat context, resulting in more accurate and actionable detection outcomes.

By focusing on lightweight deployment, explainability, and real-time detection capabilities, the proposed solution addresses key limitations of existing enterprise-grade tools. It provides SMEs with an accessible yet advanced mechanism for detecting and investigating insider threats, delivering enhanced forensic visibility and intelligent decision support without the need for complex infrastructure or extensive security expertise.

1.1 Background & Literature survey

The increasing sophistication of cyber threats has necessitated a transition from traditional signature-based detection systems toward intelligent, behavior-driven security mechanisms. Among the various threat categories, insider threats remain one of the most challenging to detect due to their inherent nature of exploiting legitimate access privileges. Unlike external attackers, insiders operate within trusted boundaries, making their activities difficult to distinguish from normal user behavior. This challenge is further amplified in Small and Medium Enterprises (SMEs), where limited cybersecurity resources and lack of advanced monitoring capabilities increase vulnerability to such attacks.

Conventional security mechanisms, including Security Information and Event Management (SIEM) and Endpoint Detection and Response (EDR) systems, primarily rely on rule-based correlation and signature matching techniques. While effective in detecting known threats, these approaches fail to identify previously unseen attack patterns, particularly those involving subtle behavioral deviations such as lateral movement, abnormal access timing, or privilege misuse. Research has demonstrated that rule-based systems exhibit high false positive rates and lack adaptability to evolving threat landscapes, limiting their effectiveness in real-time detection scenarios [2].

To overcome these limitations, machine learning-based anomaly detection has emerged as a promising alternative. Unlike supervised learning approaches, which require labeled datasets, unsupervised techniques are capable of modeling normal behavior patterns and identifying deviations without prior knowledge of attack signatures. Among these, the Isolation Forest algorithm has gained significant attention due to its efficiency in handling high-dimensional data and its ability to isolate anomalies based on recursive partitioning [5].

Let the behavioral dataset collected from endpoint telemetry be defined as

$$X = \{x_1, x_2, x_3, \dots, x_n\}, x_i \in \mathbb{R}^d$$

represents multidimensional behavioral features such as process activity, login patterns, and file access frequency. The Isolation Forest model evaluates anomalies based on the expected path length required to isolate a data point, with the anomaly score defined as

$$s_{\{IF\}}(x) = 2^{-\frac{E(h(x))}{c(n)}}$$

where $E(h(x))$ is the expected path length and $c(n)$ is the normalization factor for a dataset of size n . Data points with shorter path lengths are more likely to be anomalies, as they can be isolated more easily within the feature space [5].

While Isolation Forest effectively detects point anomalies, it lacks the ability to capture temporal dependencies in user behavior. Insider threats often manifest as sequences of actions rather than isolated events. For example, a sequence involving abnormal login timing followed by large-scale file access may indicate malicious intent, even if individual actions appear benign.

To address this limitation, sequence-based models such as Long Short-Term Memory (LSTM) networks have been widely adopted in behavioral anomaly detection. LSTM models are capable of learning temporal dependencies by maintaining internal memory states, allowing them to identify deviations in sequential patterns. Behavioral data can be modeled as a time series [6]

$$S_t = \{x_{t-k}, x_{t-k+1}, \dots, x_t\}$$

where temporal relationships between events are preserved. The anomaly score is computed based on reconstruction or prediction error,

$$S_{LSTM}(t) = \|x_t - \hat{x}\|^2$$

Where $\hat{x}_t$ represents the predicted behavior at time t . Larger deviations indicate anomalous sequences. [3]

Recent studies emphasize that combining multiple detection mechanisms significantly improves detection performance. Hybrid models that integrate statistical anomaly detection with temporal sequence learning provide a more comprehensive understanding of user behavior, enabling detection of both instantaneous and long-term anomalies.

However, machine learning models alone are insufficient for accurate threat detection due to their lack of contextual awareness. An anomaly in behavior does not necessarily imply malicious intent; therefore, contextual enrichment is required to reduce false positives and improve detection confidence. This has led to the integration of Cyber Threat Intelligence (CTI) into anomaly detection systems.

Threat intelligence platforms such as URLHaus, ThreatFOX and MalwareBazaar provide real-time information on malicious indicators, including IP addresses, domains, and malware signatures. By correlating internal behavioral anomalies with external Indicators of Compromise (IoCs), systems can differentiate between benign anomalies and actual threats. This contextual enrichment can be mathematically represented as a risk function,

$$C(x) = w_1 T(x) + w_2 U(x)$$

where $T(x)$ represents threat intelligence correlation and $U(x)$ represents user-related contextual features such as privilege level. The integration of CTI significantly enhances detection accuracy and reduces false positives by providing external validation for anomalous activities [7] [4].

Another critical challenge in ML-based detection systems is the lack of interpretability. Most models operate as black boxes, making it difficult for security analysts to understand the reasoning behind detected anomalies. To address this, Explainable Artificial Intelligence (XAI) techniques such as SHAP (Shapley Additive Explanations) have been introduced. SHAP assigns importance values to each feature based on its contribution to the model output,

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Where ϕ_i represents the contribution of feature i . This allows analysts to interpret why a particular behavior was flagged as anomalous, improving trust and usability of the system.

In addition to detection accuracy and explainability, modern cybersecurity systems must ensure forensic integrity. Insider threats often require post-incident investigation, making it essential to maintain tamper-proof logs. Techniques such as cryptographic hash chaining ensure that any modification to stored logs can be detected. This aligns with the requirements outlined in the Topic Assessment Form, which emphasizes tamper-proof forensic logging and secure evidence handling. [8]

Based on the limitations identified in existing approaches and the advancements in machine learning, threat intelligence, and explainability, this research focuses specifically on developing an AI-driven behavioral detection engine that integrates:

- Unsupervised anomaly detection (Isolation Forest)
- Temporal sequence modeling (LSTM)
- Threat intelligence enrichment
- Explainable AI mechanisms

This component forms the core analytical engine of the ForLens system, enabling accurate, interpretable, and real-time detection of insider threats in SME environments.

1.2 Research GAP

Despite significant advancements in enterprise cybersecurity platforms, existing detection systems such as Security Information and Event Management (SIEM), User and Entity Behavior Analytics (UEBA), and Extended Detection and Response (XDR) solutions continue to exhibit fundamental limitations in effectively identifying insider threats. These limitations are particularly evident when evaluated against key performance dimensions such as detection accuracy, contextual awareness, temporal modeling capability, explainability, and deployment feasibility in Small and Medium Enterprise (SME) environments.

One of the most critical gaps in existing systems is the over-reliance on rule-based and signature-driven detection mechanisms. SIEM platforms such as Splunk Enterprise Security, IBM QRadar, and Microsoft Sentinel primarily depend on predefined correlation rules and known attack signatures. While these approaches are effective for identifying known threats, they lack the ability to detect previously unseen or evolving insider attack patterns. According to the Verizon 2024 Data Breach Investigations Report, a significant proportion of breaches involve human elements, including misuse of privileges and social engineering, which often evade rule-based detection due to their subtle and context-dependent nature [1]. This results in high false negative rates, particularly for “low-and-slow” insider attacks where malicious behavior unfolds gradually over time [9] [10] [11].

In terms of detection performance metrics, traditional SIEM systems demonstrate limitations in both precision and recall. High volumes of alerts generated from static rules lead to elevated false positive rates, often exceeding acceptable operational thresholds, thereby overwhelming security analysts and reducing response efficiency. In contrast, anomaly detection techniques

such as Isolation Forest have been shown to effectively identify outliers in high-dimensional behavioral data with improved detection sensitivity [5]. However, commercial systems rarely implement such advanced models in an accessible and configurable manner.

Another significant limitation is the lack of temporal behavior modeling in existing solutions. Insider threats frequently manifest as sequences of events rather than isolated anomalies. For instance, an attack may involve a sequence of after-hours logins, privilege escalation, and gradual data exfiltration. While UEBA platforms attempt to model user behavior, they often rely on simplistic statistical baselines and fail to capture long-term dependencies in event sequences. Research demonstrates that sequence-based models such as Long Short-Term Memory (LSTM) networks significantly outperform traditional methods in detecting anomalies in time-series data by learning temporal dependencies and sequential patterns [3]. The absence of such models in most commercial solutions creates a critical gap in detecting complex insider attack scenarios.

A further limitation lies in the lack of integrated, real-time threat intelligence correlation within anomaly detection pipelines. Although modern SIEM and XDR platforms support threat intelligence ingestion, this process is typically fragmented, requiring external connectors, manual configuration, or paid subscriptions. As a result, internal behavioral anomalies are rarely correlated in real time with external Indicators of Compromise (IoCs). This disconnect reduces detection confidence and increases false positive rates. Open-source threat intelligence platforms such as URLHaus, ThreatFox, and MalwareBazaar provide continuously updated datasets of malicious URLs, file hashes, and indicators; however, these sources are not seamlessly integrated into most enterprise detection systems [4]. Consequently, existing solutions fail to leverage the full potential of contextual threat enrichment.

From a contextual risk modeling perspective, existing systems also lack multi-dimensional analysis capabilities. Most platforms generate alerts based on isolated metrics such as anomaly scores or rule matches, without incorporating additional contextual factors such as user privilege levels, threat intelligence correlation, or behavioral risk indicators. This leads to suboptimal prioritization of alerts and inefficient incident response workflows. In contrast, a context-aware anomaly detection approach that integrates behavioral, intelligence, and user-specific factors can significantly enhance detection accuracy and reduce false positives.

Another critical gap is the lack of explainability in machine learning-driven detections. Many modern security platforms incorporate proprietary machine learning models; however, these models often operate as black boxes, providing limited insight into the reasoning behind detected anomalies. This lack of transparency reduces analyst trust and complicates incident investigation processes. Explainable AI techniques such as SHAP enable feature-level attribution, allowing analysts to understand the contribution of individual variables to anomaly scores [8]. Despite their proven effectiveness, such techniques are not widely implemented in commercial detection systems in a practical and interpretable manner.

In addition to technical limitations, deployment feasibility and scalability constraints further highlight the gap between existing solutions and SME requirements. Enterprise-grade SIEM and XDR platforms require substantial infrastructure, including high-performance servers, distributed storage systems, and dedicated security teams. According to industry studies, SMEs often lack the financial and technical capacity to deploy and maintain such systems, resulting in limited adoption and increased exposure to insider threats [1]. Even open-source alternatives,

while cost-effective, demand significant expertise for configuration and tuning, making them unsuitable for lightweight, real-time deployments.

From a research and reproducibility perspective, commercial solutions present additional challenges due to their closed architectures and proprietary algorithms. This lack of transparency prevents independent validation, benchmarking, and experimentation, limiting their applicability in academic research and innovation. In contrast, research-driven approaches emphasize reproducibility, openness, and adaptability, which are essential for advancing insider threat detection methodologies.

Based on the analysis across these dimensions, it is evident that existing solutions fail to provide a unified, explainable, and context-aware anomaly detection framework that integrates advanced machine learning techniques with real-time threat intelligence in a lightweight and scalable manner.

To address these gaps, this research focuses on the development of a specialized AI-driven behavioral detection component that combines

- **Unsupervised anomaly detection (Isolation Forest)** for identifying rare behavioral deviations
- **Temporal sequence modeling (LSTM / Autoencoder-based methods)** for capturing sequential attack patterns
- **Real-time threat intelligence enrichment** using open-source platforms (URLHaus, ThreatFox, MalwareBazaar)
- **Context-aware risk scoring mechanisms** integrating behavioral, intelligence, and user-specific factors
- **Explainable AI techniques (SHAP)** for transparent and interpretable anomaly detection

This component is designed to operate as a lightweight yet powerful analytical engine, addressing the critical limitations of existing SIEM, UEBA, and XDR systems while providing improved detection accuracy, reduced false positives, enhanced interpretability, and practical deployability within SME environments.

Metric / Capability	SIEM / XDR Systems	Proposed ML + TI Component
Detection Type	Primarily rule-based and signature-driven detection (static correlation rules) [1]	Unsupervised ML-based anomaly detection (Isolation Forest, Autoencoder) capable of identifying unknown patterns [5]
Unknown Threat Detection	Limited detection of zero-day / insider threats; SIEM struggles with “low-and-slow” attacks [1]	High capability due to anomaly-based detection without labeled data [5]
False Positive Rate	High alert volumes; SOC analysts spend ~20–30% time investigating false positives [1]	Reduced through TI correlation + behavioral context (multi-factor scoring) [4]

Temporal Modeling	Limited or coarse-grained behavior baselining (UEBA statistical models) [1]	Advanced sequence modeling using LSTM/Autoencoder capturing time-series behavior [3]
Threat Intelligence Integration	Often manual or connector-based; fragmented ingestion of feeds [1]	Automated enrichment using real-time OSINT feeds (URLHaus, ThreatFox, MalwareBazaar) [4]
Context-Aware Risk Scoring	Mostly rule-triggered alerts or static scoring models [1]	Multi-factor risk scoring combining behavior + TI + user context
Explainability	Limited transparency; proprietary models provide minimal reasoning [6]	SHAP-based explainable outputs provide feature-level reasoning [8]
SME Suitability	High cost, complex infrastructure; SIEM adoption low among SMEs [1]	Lightweight architecture designed for low-resource environments
Deployment Complexity	Requires dedicated SOC teams, infrastructure, and tuning [1]	Simple deployment with minimal configuration and scalable ML pipeline

Table 1: Comparison of SIEM/XDR Systems and Proposed ML + TI Component

1.3 Research Problem

Insider threats remain one of the most difficult classes of cybersecurity incidents to detect because they originate from users with legitimate access and often manifest through subtle, context-dependent behavioral deviations. Unlike external attacks that can be identified through signatures or known indicators, insider threats typically unfold as low-frequency, high-impact sequences of actions for example, unusual login times followed by incremental data access and eventual exfiltration. Industry analyses consistently show that a large share of breaches involve the human element (misuse, error, or credential compromise) and that such incidents frequently evade traditional controls and persist for extended periods before detection [1] [12] .

Current enterprise detection stacks SIEM, UEBA, and XDR are constrained by architectural and analytical limitations that reduce their effectiveness against these patterns. First, they rely heavily on rule-based correlation and signature-driven logic, which are inherently reactive and insufficient for detecting novel or evolving behaviors. Studies and industry reports note that rule-centric systems generate substantial alert volumes, contributing to alert fatigue and non-trivial false positive rates, while still missing stealthy insider activity that does not match predefined rules [1] [13]. Second, while UEBA attempts to model behavior, many implementations depend on coarse statistical baselines rather than robust temporal models, limiting their ability to capture sequential dependencies across events.

From a modeling perspective, effective insider threat detection requires capturing both point anomalies (rare events) and temporal anomalies (suspicious sequences). Unsupervised methods such as Isolation Forest can identify outliers in high-dimensional behavioral data without labeled examples [5], while sequence models such as LSTM learn dependencies across time and have demonstrated improved performance on log-based anomaly detection tasks [3]. However, these techniques are not systematically integrated, tunable, or explainable within

most commercial platforms, and when present, are often opaque due to proprietary implementations.

A further limitation is the insufficient, fragmented integration of threat intelligence into the detection loop. Effective decision-making requires correlating internal anomalies with external Indicators of Compromise (IoCs) such as malicious URLs, domains, file hashes. Although SIEM/XDR products can ingest threat feeds, integration is commonly connector-driven, delayed, or subscription-bound, limiting real-time enrichment and correlation. Meanwhile, open-source intelligence sources such as URLHaus, ThreatFox, and MalwareBazaar provide continuously updated, high-quality indicators, yet are rarely fused directly and automatically into anomaly scoring pipelines [4]. The absence of this real-time enrichment reduces detection confidence and contributes to both false positives (benign anomalies flagged) and false negatives (malicious anomalies lacking context).

Another critical barrier is lack of explainability. Security analysts must understand why an alert was raised to triage and respond effectively. Many existing systems provide limited reasoning such as rule IDs or aggregate risk scores, while ML-driven detections often behave as black boxes. Explainable AI methods such as SHAP offer feature-level attribution for model outputs, enabling transparent, per-alert explanations; however, such techniques are seldom operationalized in mainstream security tools [8].

Operational constraints further exacerbate the problem. Enterprise SIEM/XDR deployments demand significant infrastructure, licensing, and specialized expertise, which are often beyond the reach of SMEs [12] [13]. This creates a gap where organizations most vulnerable to insider threats lack access to effective, real-time, behavior-aware detection.

Despite the availability of advanced security tools, accurately detecting insider threats remains a significant challenge, particularly for small and medium-sized enterprises (SMEs) with limited computational resources. Traditional anomaly detection approaches often rely on static rules or isolated behavioral indicators, which can lead to high false positive rates and an inability to detect previously unseen attack patterns.

This research addresses the challenge of designing a lightweight and practical anomaly detection component that not only identifies unusual user behavior at a single point in time, but also captures evolving patterns over time. In addition, it explores how integrating real-time insights from open-source threat intelligence sources such as URLHaus, ThreatFox, and MalwareBazaar can enhance detection accuracy by providing contextual awareness of emerging threats.

The core problem, therefore, is how to effectively combine behavioral machine learning with external threat intelligence in a way that reduces false alerts, improves the detection of novel or stealthy insider activities, and remains efficient enough to be deployed in resource-constrained environments without sacrificing interpretability or transparency.

Addressing this problem requires a unified analytical component that

- (i) Models behavioral baselines and temporal sequences.
- (ii) Performs real-time TI enrichment and correlation.
- (iii) Produces interpretable explanations for each alert.
- (iv) Operates with low computational overhead suitable for SME environments.

The proposed component in this research is designed to meet these requirements by integrating Isolation Forest and sequence models (LSTM/Autoencoder) with automated abuse.ch TI feeds and SHAP-based explainability into a single, cohesive detection pipeline.

1.4 Research Objectives

1.4.1 Main Objective

The primary objective of this research is to design and develop a lightweight, explainable, and context-aware behavioral anomaly detection component capable of identifying threats in real time by integrating unsupervised machine learning techniques with dynamic threat intelligence enrichment. The proposed component aims to overcome the limitations of traditional rule-based detection systems by improving detection accuracy, reducing false positives, enabling interpretability, and ensuring practical deployment within resource-constrained Small and Medium Enterprise (SME) environments.

1.4.2 Specific Objectives

To achieve the overall goal of developing a lightweight, explainable, and context-aware threat detection component, this research focuses on a set of specific objectives that directly address the limitations identified in existing SIEM and XDR systems. These objectives are centered around the design and implementation of an intelligent behavioral analysis engine that integrates machine learning, temporal modeling, and real-time threat intelligence. Each objective contributes to improving detection accuracy, reducing false positives, and enabling practical deployment within SME environments.

1. Development of Behavioral Anomaly Detection Using Unsupervised Machine Learning

This objective focuses on designing an anomaly detection mechanism capable of identifying deviations in user and system behavior without relying on labeled datasets. Given the dynamic and unpredictable nature of insider threats, unsupervised learning approaches are employed to establish a baseline of normal behavior and detect anomalies that may indicate malicious activity.

- Implementation of Isolation Forest for detecting statistical outliers
- Modeling of high-dimensional behavioral features (process, login, file activity)
- Detection of unknown and zero-day insider threats without predefined signatures

2. Incorporation of Temporal Behavior Analysis for Sequential Threat Detection

Insider threats often occur as sequences of actions rather than isolated events. This objective addresses the need to capture temporal dependencies in user activity by integrating sequence-based learning techniques into the detection process.

- Use of LSTM / Autoencoder-based models for time-series analysis
- Detection of multi-stage and “low-and-slow” insider attacks
- Analysis of behavioral sequences instead of individual events

3. Integration of Real-Time Threat Intelligence for Contextual Enrichment

This objective aims to enhance anomaly detection by incorporating external threat intelligence data. By correlating detected anomalies with known malicious indicators, the system can improve detection confidence and reduce false positives.

- Integration of URLHaus, ThreatFox, and MalwareBazaar as primary TI sources
- Enrichment of anomalies with IoCs (malicious URLs, IPs, file hashes)
- Real-time correlation between internal behavior and external threat data

4. Development of a Context-Aware Multi-Factor Risk Scoring Mechanism

To improve decision-making and alert prioritization, this objective focuses on combining multiple factors into a unified risk scoring model.

- Combination of anomaly scores + threat intelligence + user context
- Incorporation of user privilege levels and behavioral patterns
- Reduction of alert noise through intelligent prioritization

5. Implementation of Explainable AI for Transparent Detection

This objective addresses the lack of interpretability in machine learning models by integrating explainable AI techniques to provide clear reasoning behind detection decisions.

- Use of SHAP (Shapley Additive Explanations) for feature importance
- Generation of human-readable explanations for anomalies
- Improvement of analyst trust and investigation efficiency

6. Design of a Lightweight and Scalable Architecture for SME Environments

This objective ensures that the proposed solution remains practical and deployable in environments with limited resources, particularly SMEs.

- Development of a lightweight detection component
- Minimization of computational and infrastructure requirements
- Elimination of dependency on heavy SIEM/XDR systems

7. Performance Evaluation Using Quantitative Metrics

To validate the effectiveness of the proposed component, this objective focuses on evaluating system performance using measurable criteria.

- Evaluation using accuracy, precision, recall, and F1-score
- Measurement of false positive rate and detection latency
- Comparison with traditional rule-based detection approaches

2. Methodology

2.1 Understanding Key Pillars of the Research Domain

The development of an effective threat detection component requires a comprehensive understanding of several interrelated research domains, including machine learning-based anomaly detection, temporal behavior modeling, cyber threat intelligence integration, explainable artificial intelligence, and context-aware risk analysis. These pillars collectively form the foundation for designing a detection system capable of identifying complex and evolving insider threats in real time.

A fundamental pillar of this research is machine learning-based anomaly detection, which enables the identification of abnormal behavior without relying on predefined rules or labeled attack data. Traditional detection mechanisms, such as rule-based SIEM systems, are limited in their ability to detect unknown or evolving threats, particularly those involving legitimate user credentials. Industry evidence highlights that a significant proportion of cyber incidents involve misuse of authorized access, which often bypasses signature-based detection systems [1]. To address this limitation, unsupervised learning techniques are employed to model baseline user behavior and detect deviations. In this research, the Isolation Forest algorithm is utilized due to its efficiency in handling high-dimensional datasets and its capability to isolate anomalies through recursive partitioning [5]. This approach allows the system to identify rare and suspicious activities without prior knowledge of attack patterns.

Another critical pillar is temporal behavior modeling, which addresses the sequential nature of insider threats. Unlike external attacks, insider activities often unfold over time as a series of seemingly benign actions. Detecting such patterns requires the ability to analyze event sequences rather than individual data points. Research has demonstrated that sequence-based models, such as Long Short-Term Memory (LSTM) networks, significantly enhance anomaly detection by capturing temporal dependencies within system logs and user activities [3]. By incorporating temporal analysis, the system can identify complex attack patterns such as gradual data exfiltration, repeated unauthorized access, or privilege escalation over time. This temporal perspective is essential for detecting “low-and-slow” insider threats that are otherwise difficult to identify.

The integration of cyber threat intelligence (CTI) represents another key pillar of the research domain. While anomaly detection techniques can identify deviations in behavior, they lack contextual awareness regarding whether such deviations are malicious. To overcome this limitation, external threat intelligence is incorporated to provide real-time context. This research integrates open-source intelligence platforms, including URLHaus, ThreatFox, and MalwareBazaar, which provide continuously updated Indicators of Compromise (IoCs) such as malicious IP addresses, domains, and file hashes [4]. By correlating internal anomalies with external threat intelligence, the system improves detection accuracy, reduces false positives, and enhances the overall reliability of alerts.

In addition to detection and enrichment, the interpretability of machine learning models is a critical requirement in cybersecurity applications. Many advanced detection systems rely on complex algorithms that operate as black boxes, making it difficult for analysts to understand the reasoning behind alerts. This lack of transparency limits trust and hinders effective incident response. To address this challenge, the research incorporates explainable artificial intelligence

techniques, specifically SHAP (Shapley Additive Explanations). SHAP provides feature-level attribution, enabling analysts to understand how individual features contribute to anomaly detection decisions [8]. This enhances the usability of the system by providing clear, interpretable insights into detected threats.

Another important pillar is context-aware risk analysis, which involves combining multiple sources of information to generate meaningful threat assessments. Traditional systems often rely on isolated indicators, such as rule triggers or anomaly scores, which can result in inaccurate prioritization of alerts. In contrast, a context-aware approach integrates behavioral anomalies, threat intelligence correlations, and user-specific contextual factors, such as privilege levels and activity history. This multi-dimensional analysis improves decision-making and ensures that alerts are prioritized based on their actual risk level.

Furthermore, the research emphasizes the importance of using realistic datasets for model training and evaluation. The system utilizes the CERT Insider Threat Dataset developed by Carnegie Mellon University's Software Engineering Institute, which provides synthetic but realistic user behavior logs, including logon events, file access activities, and system interactions. This dataset is widely used in insider threat research due to its comprehensive representation of both normal and malicious behaviors [14]. The use of such a dataset ensures that the developed models are trained on realistic scenarios, thereby improving their applicability in real-world environments.

Finally, the need for lightweight and scalable system design is a crucial consideration, particularly for Small and Medium Enterprises (SMEs). Many existing enterprise security solutions require substantial infrastructure and expertise, limiting their accessibility. Reports indicate that SMEs face significant challenges in adopting advanced cybersecurity solutions due to resource constraints [13]. Therefore, this research focuses on designing a detection component that balances performance and efficiency, ensuring that advanced anomaly detection and threat intelligence capabilities can be deployed with minimal computational overhead.

In summary, the research domain is built upon five key pillars; machine learning-based anomaly detection, temporal behavior modeling, threat intelligence integration, explainable artificial intelligence, and context-aware risk analysis. These pillars collectively enable the development of an intelligent, efficient, and interpretable insider threat detection component that addresses the limitations of traditional security systems while remaining practical for real-world deployment.

2.1.1 Machine Learning-Based Anomaly Detection

Machine learning-based anomaly detection forms the foundational analytical mechanism of the proposed threat detection component. Unlike traditional rule-based security systems, which rely on predefined signatures and static thresholds, anomaly detection techniques enable the identification of previously unseen or evolving threats by modeling normal behavioral patterns and detecting deviations. This capability is particularly critical in insider threat scenarios, where malicious actions are often subtle, context-dependent, and indistinguishable from legitimate user activity.

In cybersecurity, anomaly detection is commonly categorized into supervised, semi-supervised, and unsupervised approaches. However, threat detection presents a unique

challenge due to the scarcity of labeled malicious data. Real-world datasets rarely contain comprehensive and accurately labeled insider attack scenarios, making supervised learning approaches impractical. Consequently, unsupervised learning methods have gained prominence, as they can learn normal behavior distributions directly from data and identify anomalies without requiring labeled training samples [15].

In this research, anomaly detection is implemented using a **hybrid machine learning framework** that combines statistical outlier detection with deep learning-based sequence modeling. The primary components include the Isolation Forest algorithm and a Long Short-Term Memory (LSTM) Autoencoder, both of which are integrated into a unified scoring mechanism to improve detection accuracy and robustness.

The Isolation Forest algorithm is employed as the core statistical anomaly detection model. It operates on the principle that anomalous data points are more susceptible to isolation than normal observations. By recursively partitioning the data space using random decision trees, the algorithm computes the path length required to isolate each data point. Anomalies, being rare and distinct, typically require fewer splits and therefore have shorter path lengths. The anomaly score is mathematically defined as,

$$S_{\{IF\}}(x) = 2^{-\frac{E(h(x))}{c(n)}}$$

where $E(h(x))$ represents the expected path length for a data point x , and $c(n)$ is a normalization factor dependent on the dataset size n . A score closer to 1 indicates a higher likelihood of anomaly, while values closer to 0 represent normal behavior [5]. The effectiveness of Isolation Forest in high-dimensional datasets and its low computational overhead make it particularly suitable for real-time cybersecurity applications.

While Isolation Forest effectively detects point anomalies, it does not capture temporal dependencies inherent in user behavior. Insider threats often manifest as sequences of actions over time rather than isolated events. To address this limitation, this research incorporates a **Long Short-Term Memory (LSTM) Autoencoder**, a deep learning model designed to learn temporal patterns in sequential data.

The LSTM Autoencoder is trained to reconstruct normal behavioral sequences by encoding input sequences into a latent representation and then decoding them back to their original form. The reconstruction error serves as an anomaly indicator, as the model is unable to accurately reconstruct sequences that deviate from learned normal patterns. The anomaly score is computed as the mean squared error between the original and reconstructed sequences:

$$S_{LSTM}(t) = \|x_t - \hat{x}\|^2$$

Where x_t represents the observed sequence and $\hat{x}$ represents the reconstructed sequence. Higher reconstruction errors indicate a higher likelihood of anomalous behavior [3]. This approach enables the detection of complex temporal patterns such as gradual privilege escalation, abnormal session behavior, and multi-stage insider attacks.

To further enhance detection performance, the outputs of the Isolation Forest and LSTM Autoencoder are combined using a hybrid anomaly scoring mechanism. This approach leverages the strengths of both models: the ability of Isolation Forest to detect statistical outliers

and the capability of LSTM to capture temporal dependencies. The hybrid score is computed as a weighted combination of individual model outputs:

$$S_{hybrid} = \alpha \cdot S_{IF} + \beta \cdot S_{LSTM}$$

Where S_{IF} and S_{LSTM} represent the anomaly scores from the Isolation Forest and LSTM models, respectively, and α and β are weighting coefficients that control the contribution of each model. This fusion mechanism improves robustness by reducing model-specific biases and capturing a broader range of anomalous behaviors.

The anomaly detection models are trained and evaluated using the CERT Insider Threat Dataset developed by Carnegie Mellon University’s Software Engineering Institute. This dataset provides realistic synthetic user activity logs, including logon events, file access records, and system interactions, enabling the simulation of insider threat scenarios. The dataset is widely used in academic research due to its comprehensive representation of both normal and malicious behaviors [14]. By utilizing this dataset, the proposed system ensures that the learned behavioral patterns are representative of real-world enterprise environments.

Experimental analysis demonstrates that the hybrid model provides improved detection performance compared to individual models. The integration of statistical and temporal anomaly detection allows the system to identify both instantaneous anomalies and long-term behavioral deviations. Additionally, the distribution of anomaly scores reveals clear separation between normal and anomalous activities, supporting effective threshold-based classification.

Overall, machine learning-based anomaly detection in this research provides a scalable, adaptive, and data-driven approach to insider threat detection. By combining unsupervised learning techniques with temporal modeling and hybrid scoring, the proposed system overcomes the limitations of traditional rule-based methods and enables accurate detection of complex and evolving insider threats.

2.1.2 Temporal Behavior Modeling

Temporal behavior modeling is a critical component in threat detection, as malicious activities rarely occur as isolated events but rather emerge through sequences of actions distributed over time. Unlike external attacks that may exhibit immediate and identifiable signatures, insider threats typically manifest as gradual, context-dependent behavioral deviations such as repeated after-hours logins, progressive privilege misuse, or staged data exfiltration. These patterns are inherently sequential and cannot be effectively captured using static, point-in-time anomaly detection techniques. Therefore, incorporating temporal modeling into the detection framework is essential for accurately identifying complex insider attack scenarios.

Traditional anomaly detection methods, including statistical and distance-based approaches, treat observations as independent data points and ignore temporal dependencies. This limitation significantly reduces their effectiveness in cybersecurity contexts, where the order, frequency, and timing of events carry critical semantic meaning. Research has demonstrated that incorporating temporal context improves anomaly detection performance by enabling the identification of sequential dependencies and long-term behavioral patterns [15]. In particular, sequence modeling techniques based on recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM) networks, have shown strong performance in modeling time-series data and detecting anomalies in system logs and user activity streams [3].

In this research, temporal behavior modeling is implemented using a multi-layered approach that combines session-based aggregation, sliding window analysis, and deep learning-based sequence reconstruction. The first stage involves transforming raw event logs into temporally structured sequences. User activities, including login events, process executions, file access operations, and network interactions, are grouped into sessions or fixed-length time windows. This transformation allows the system to capture behavioral patterns over defined intervals rather than treating each event independently.

Formally, a temporal sequence can be represented as:

$$S_t = \{x_{t-k}, x_{t-k+1}, \dots, x_t\}$$

Where S_t represents a sequence of events within a time window of size k , and each x_i corresponds to a feature vector representing user activity at time i . This formulation enables the modeling of temporal dependencies and the detection of deviations in sequential behavior.

To analyze these sequences, the system employs a **Long Short-Term Memory (LSTM) Autoencoder**, a deep learning architecture specifically designed for sequence modeling. LSTM networks address the limitations of traditional RNNs by incorporating memory cells and gating mechanisms that allow the model to retain long-term dependencies while avoiding issues such as vanishing gradients. This makes them particularly effective for capturing patterns in sequential data where dependencies may span across extended time intervals [6].

The LSTM Autoencoder is trained on sequences representing normal user behavior. During training, the model learns to encode each input sequence into a compressed latent representation and then reconstruct it with minimal error. The reconstruction error is used as a measure of anomaly, as sequences that deviate from learned patterns result in higher reconstruction loss. The anomaly score for a sequence is computed as,

$$S_{LSTM}(t) = \frac{1}{K} \sum_{i=1}^k (x_i - \hat{x}_i)^2$$

Where x_i is the original feature vector and $\hat{x}_i$ is the reconstructed output. Higher values of $S_{LSTM}(t)$ indicate greater deviation from normal behavior and are therefore indicative of potential anomalies.

This approach enables the detection of complex temporal patterns, including,

- Gradual escalation of abnormal activity
- Repeated access to sensitive resources over time
- Deviations in user session behavior
- Multi-stage attack sequences involving coordinated actions

In addition to deep learning-based modeling, temporal behavior analysis in this research incorporates sliding window mechanisms and session-based feature aggregation. Sliding windows allow continuous monitoring of user activity streams by evaluating overlapping sequences, thereby improving sensitivity to short-term behavioral changes. Session-based

aggregation, on the other hand, captures user behavior within logical activity boundaries, such as login sessions or task-specific interactions. These techniques ensure that both short-term anomalies and long-term behavioral trends are effectively captured.

The integration of temporal modeling with statistical anomaly detection further enhances the robustness of the system. While Isolation Forest detects point, anomalies based on feature distributions, the LSTM Autoencoder captures deviations in sequential patterns. By combining these approaches within a hybrid framework, the system achieves comprehensive coverage of both instantaneous and temporal anomalies.

The effectiveness of temporal behavior modeling is validated using the CERT Insider Threat Dataset, which contains time-stamped user activity logs representing realistic enterprise environments. The dataset includes sequences of actions performed by users over extended periods, enabling the evaluation of temporal detection capabilities. Experimental results indicate that incorporating temporal modeling significantly improves the detection of insider threats, particularly those involving slow and stealthy attack patterns that are not identifiable through static analysis alone [14].

Furthermore, temporal modeling contributes to reducing false positives by providing contextual continuity. Isolated anomalies that may appear suspicious in static analysis can be validated or dismissed based on their surrounding sequence context. This improves the reliability of detection outcomes and supports more accurate risk assessment.

Temporal behavior modeling serves as a crucial enhancement to anomaly detection by enabling the analysis of sequential dependencies and long-term behavioral patterns. Through the integration of LSTM-based sequence modeling, sliding window analysis, and session-based aggregation, the proposed system effectively captures the dynamic nature of insider threats. This capability significantly improves detection accuracy and complements statistical anomaly detection methods, resulting in a more robust and context-aware security framework.

2.1.3 Threat Intelligence Integration

Threat Intelligence Integration plays a pivotal role in enhancing the effectiveness of anomaly-based insider threat detection systems by providing external contextual validation for internally detected anomalies. While machine learning-based anomaly detection techniques are capable of identifying deviations from normal behavior, they often lack the contextual understanding required to determine whether such deviations are truly malicious. This limitation frequently leads to high false positive rates, as not all anomalous behavior corresponds to security threats. Therefore, integrating cyber threat intelligence (CTI) into the detection pipeline enables a more informed and accurate decision-making process.

Cyber threat intelligence refers to the collection, analysis, and dissemination of information related to known and emerging threats, including Indicators of Compromise (IoCs) such as malicious IP addresses, domain names, URLs, and file hashes. Industry frameworks, including the Structured Threat Information Expression (STIX) and Trusted Automated Exchange of Indicator Information (TAXII), emphasize the importance of standardized threat intelligence sharing to improve situational awareness and threat detection capabilities across organizations [16]. By incorporating such intelligence into detection systems, it becomes possible to correlate internal behavioral anomalies with externally verified malicious indicators.

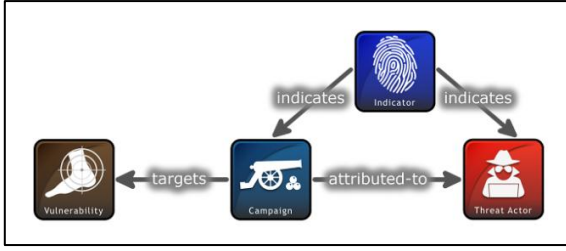


Figure 1: STIX Relationship Example

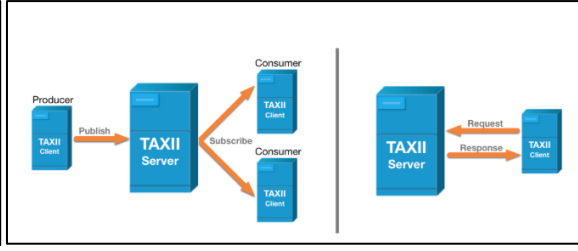


Figure 2: TAXII Collections

In this research, threat intelligence integration is implemented as a multi-stage pipeline that encompasses data acquisition, normalization, enrichment, storage, and real-time correlation. The system integrates multiple open-source intelligence sources provided by abuse.ch, specifically URLHaus, ThreatFox, and MalwareBazaar. These platforms offer continuously updated and community-driven threat intelligence datasets that include high-confidence IoCs derived from real-world malware campaigns and cyber incidents [4]. The use of multiple sources ensures comprehensive coverage of different types of threats, including network-based indicators, phishing domains, and malware signatures.

A key challenge in integrating multiple threat intelligence sources is data heterogeneity, as different platforms provide data in varying formats, structures, and levels of detail. To address this issue, the system implements a normalization layer that transforms raw threat intelligence data into a unified schema. Each IoC is standardized into a structured format containing attributes such as indicator type, value, confidence score, severity level, source, and associated metadata. This normalization process ensures consistency across data sources and facilitates efficient processing and correlation.

Following normalization, the system incorporates MITRE ATT&CK framework mapping to provide a higher level of contextual understanding. The MITRE ATT&CK framework is a widely adopted knowledge base that categorizes adversarial tactics and techniques based on real-world observations [17]. By mapping IoCs to specific ATT&CK techniques and tactics, the system enhances its ability to interpret detected threats within a broader attack context. This mapping is achieved through rule-based matching of IoC attributes and is further supported by periodic updates from official MITRE datasets, ensuring that the system remains aligned with evolving threat landscapes.

The normalized and enriched threat intelligence data is stored in a local database to support efficient querying and real-time access. This persistent storage enables the system to maintain historical intelligence data, perform fast lookups during anomaly detection, and avoid repeated API calls to external sources. Efficient indexing and upsert mechanisms ensure that the database remains up-to-date while minimizing redundancy.

The core functionality of threat intelligence integration lies in the real-time correlation of IoCs with detected anomalies. During the detection phase, relevant indicators are extracted from behavioral events, such as IP addresses from network connections, domains from HTTP requests, and email addresses from communication logs. These extracted indicators are then matched against the threat intelligence database to identify potential associations with known malicious entities.

If a match is identified, the system assigns a threat intelligence score based on factors such as the confidence level, severity rating, and frequency of occurrence across multiple sources. This

score is then incorporated into the overall risk assessment process. Formally, the contribution of threat intelligence to the final risk score can be represented as,

$$S_{TI} = \sum_{i=1}^n w_i \cdot c_i$$

where c_i represents the confidence score of the matched IoC and w_i represents the weight assigned based on its severity or source reliability. This formulation allows the system to quantify the impact of threat intelligence on the overall detection decision.

The integration of threat intelligence significantly enhances detection performance by reducing false positives and increasing detection confidence. Anomalies that are correlated with known malicious indicators are prioritized as high-risk events, while those without such correlation can be further analyzed or deprioritized. This context-aware approach aligns with modern cybersecurity practices, which emphasize the importance of combining behavioral analytics with external intelligence to improve threat detection accuracy [13].

Furthermore, the integration of threat intelligence supports attack chain analysis and incident correlation. By linking multiple IoCs associated with different stages of an attack, the system can identify patterns indicative of coordinated malicious activity. This capability is further enhanced through graph-based correlation techniques, enabling the visualization and analysis of relationships between events, indicators, and attack techniques.

The effectiveness of the threat intelligence integration approach is validated through its application to real-world datasets, including the CERT Insider Threat Dataset, where externally sourced IoCs are used to enrich internally generated anomalies. Experimental observations indicate that the inclusion of threat intelligence leads to a significant reduction in false positives and improves the prioritization of alerts, particularly in scenarios involving network-based anomalies and suspicious communications.

Threat intelligence integration serves as a critical enhancement to machine learning-based anomaly detection by providing external validation and contextual enrichment. Through the combination of multi-source data ingestion, normalization, MITRE ATT&CK mapping, persistent storage, and real-time correlation, the proposed system achieves a comprehensive and context-aware detection capability. This integration not only improves detection accuracy but also enables more effective incident analysis and response, thereby addressing key limitations of traditional anomaly detection systems.

2.1.4 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) has emerged as a critical requirement in modern cybersecurity systems, particularly those that rely on machine learning for threat detection. While advanced machine learning models provide high detection accuracy, they often operate as black-box systems, offering limited insight into the reasoning behind their predictions. In security operations, where decisions must be validated and acted upon by human analysts, the lack of interpretability significantly reduces trust and limits the practical usability of automated detection systems. Therefore, integrating explainability into anomaly detection frameworks is essential for enabling transparency, accountability, and effective incident response.

In the context of insider threat detection, explainability is particularly important due to the subtle and context-dependent nature of malicious behavior. An anomaly score alone is insufficient for determining whether an observed activity represents a genuine threat. Analysts require detailed explanations that highlight the contributing factors, contextual relevance, and potential attack implications of detected anomalies. Research in explainable AI emphasizes the importance of interpretable models and post-hoc explanation techniques to bridge the gap between model predictions and human understanding [18].

This research adopts a multi-layered explainability framework that extends beyond traditional feature attribution methods to provide comprehensive, context-aware explanations. The proposed XAI architecture integrates model-level interpretation, behavioral deviation analysis, threat intelligence reasoning, and decision traceability into a unified explanation pipeline.

At the core of the explainability framework is the use of SHAP (Shapley Additive Explanations), a state-of-the-art method for interpreting machine learning predictions. SHAP is based on cooperative game theory and computes the contribution of each feature to a model's output by considering all possible feature combinations. The SHAP value for a feature is defined as,

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

where ϕ_i represents the contribution of feature i , S is a subset of features, and f is the model prediction function [8]. This formulation ensures a fair and consistent attribution of feature importance, making SHAP particularly suitable for explaining complex models such as Isolation Forest and neural networks.

While SHAP provides robust feature-level explanations, it may not always be computationally efficient for large datasets or complex models. To address this limitation, the system incorporates LIME (Local Interpretable Model-agnostic Explanations) as a fallback mechanism. LIME approximates the behavior of complex models locally using interpretable surrogate models, enabling efficient generation of explanations for individual predictions [19]. The combination of SHAP and LIME ensures both accuracy and scalability in the explanation process.

Beyond model-level explanations, the proposed framework introduces behavioral deviation analysis, which provides insights into how user activity deviates from established baselines. This layer analyzes key behavioral features, such as login times, file access patterns, and network interactions, to identify deviations from normal activity profiles. For example, after-hours access, unusual data transfer volumes, or interactions with previously unseen domains are highlighted as contributing factors to anomaly detection. This approach bridges the gap between abstract model outputs and real-world behavioral context.

A significant enhancement in this research is the integration of threat intelligence-based explainability, which contextualizes anomalies using external intelligence sources. When an anomaly is associated with a known malicious indicator, such as a suspicious IP address or domain, the system provides detailed reasoning that includes the source of the intelligence, confidence level, and associated threat categories. This enables analysts to understand not only that an anomaly exists, but also why it is considered malicious based on external evidence.

The framework further incorporates MITRE ATT&CK mapping, which links detected anomalies and associated IoCs to known adversarial tactics and techniques. By mapping events to specific ATT&CK techniques, the system provides a structured understanding of potential attack scenarios, enabling analysts to interpret anomalies within the context of known threat behaviors [17]. This mapping enhances situational awareness and supports more effective threat hunting and incident response.

Another critical component of the XAI framework is the decision trace mechanism, which provides full transparency into the detection process. This mechanism records intermediate outputs from all stages of the detection pipeline, including anomaly scores from Isolation Forest and LSTM models, hybrid scoring contributions, threshold evaluations, and threat intelligence correlations. By presenting a complete trace of how a decision was reached, the system ensures auditability and supports forensic analysis.

To improve usability, the system includes a narrative generation layer that converts technical explanations into human-readable summaries. This layer synthesizes information from model outputs, behavioral analysis, and threat intelligence to produce concise descriptions of detected anomalies. These narratives enable security analysts to quickly understand the nature of a threat without requiring deep technical knowledge of the underlying models.

The integration of these components results in a comprehensive explainability framework that operates at multiple levels, including model interpretation, behavioral context, intelligence correlation, and decision transparency. This multi-layered approach aligns with recent advancements in explainable AI, which emphasize the need for holistic explanations that combine multiple perspectives rather than relying on single-method interpretations [20].

The effectiveness of the proposed XAI framework is demonstrated through its ability to improve analyst trust, reduce investigation time, and enhance decision-making accuracy. By providing clear and interpretable explanations, the system enables analysts to validate alerts, understand underlying causes, and take appropriate actions with greater confidence. Furthermore, the inclusion of explainability supports compliance with emerging regulatory requirements that mandate transparency in automated decision-making systems.

Explainable artificial intelligence serves as a critical enabler for the practical deployment of machine learning-based security systems. Through the integration of SHAP, LIME, behavioral analysis, threat intelligence reasoning, MITRE mapping, decision traceability, and narrative generation, the proposed framework provides a comprehensive and interpretable approach to insider threat detection. This not only enhances the reliability of the system but also bridges the gap between automated detection and human-centric security operations.